GRAPH THEORY: HOMEWORK 2 BOOK SECTION 1.2.

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1. Suppose that every vertex of a simple graph G has degree at least 3. Prove that G has a cycle of even length. (Hint: Consider a maximal path $P = v_1 v_2 \cdots v_k$)

Proof. $\square$

2. Prove that a bipartite graph has a unique bipartition (except for interchanging the two partite sets) if and only if it is connected.

Proof. $\square$

3. Let v be a cut-vertex of a simple graph G. Prove that $\overline{G} - v$ is connected.

Proof. $\square$

4. Let G be a simple graph having no isolated vertex and no induced subgraph with exactly two edges. Prove that G is a complete graph.

Proof. $\square$

5. Let G be a connected simple graph not having P_4, or C_3 as an induced sub-graph. Prove that G is a biclique (complete bipartite graph). (Hint: Recall that the distance $d_G(u, v)$ between vertices u and v in G is the shortest path between u and v [assuming that G is connected]. Fix an arbitrary vertex $v \in V(G)$. Partition $V(G)$ using distances to v by letting $L_i = \{u \in V(G) : d_g(v, u) = i\}$ for $i \geq 0$. Why must $L_i =$ for $i \geq 3$. Why must G be complete bipartite?)

Proof. $\square$

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